Docker Containerization

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Aim

To install Docker on Windows Subsystem for Linux (WSL), create and manage containers, and execute a Python environment inside a Docker container.

Objective

- Install Docker Engine in WSL.
- Verify Docker functionality.
- Run lightweight containers.
- Deploy a web server container.
- Execute Python inside a containerized environment.

System Requirements

- Windows 10/11 with WSL enabled
- Ubuntu (WSL distribution)
- Docker Engine
- Internet connectivity

Procedure

Step 1 – Install Docker

Docker packages and dependencies were installed using the apt package manager.

Command used:

sudo apt update

sudo apt install docker.io

Step 2 – Start Docker Service

The Docker service was started and verified.

sudo systemctl start docker

docker –version

Step 3 – Test Docker with BusyBox

A simple container was executed to confirm Docker is working.

sudo docker run busybox echo "Docker is working successfully"

Step 4 – Run Alpine Linux Container

An interactive Alpine container was launched.

docker run -it alpine sh

This provided a minimal Linux environment inside Docker.

```
kavip@KaviPrakash:~$ docker run -it --name alpine_test alpine sh
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
589002ba0eae: Pull complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
/ # uname -a
Linux 7e91307e5ac4 5.15.167.4-microsoft-standard-WSL2 #1 SMP Tue Nov 5 00:21:55 UTC 2024 x86_64 Linux
/ # ls /
bin    dev    etc    home   lib    media  mnt    opt    proc   root   run    sbin   srv    sys    tmp    usr    var
/ #
```

Step 5 – Deploy Apache Web Server

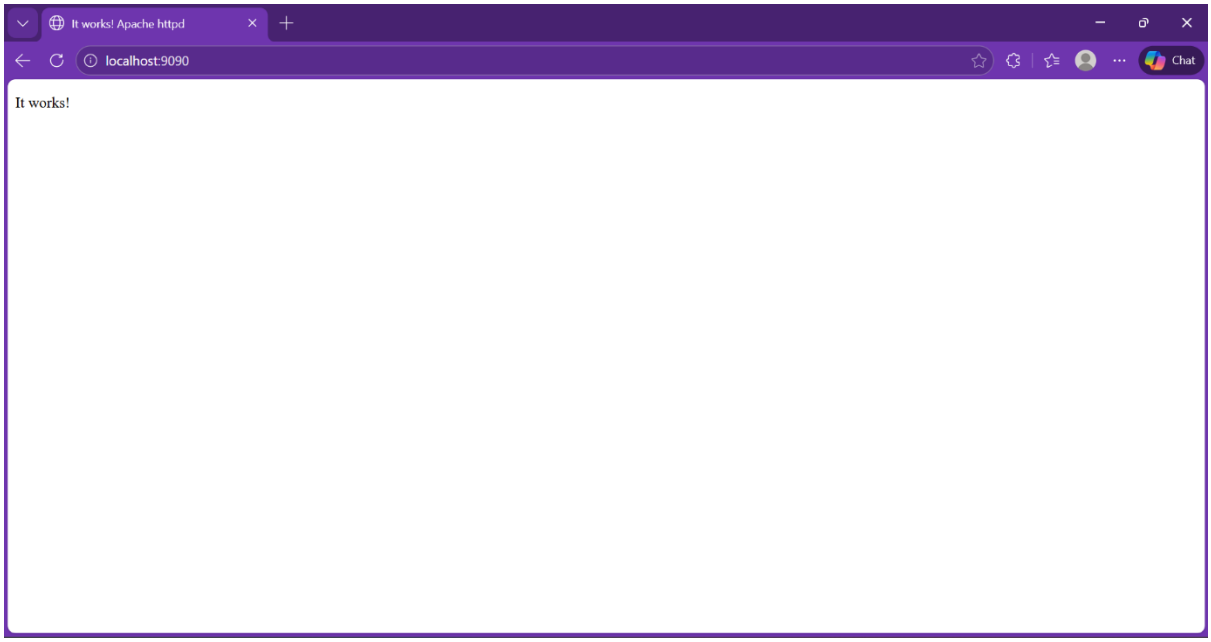
An Apache container was created and mapped to port 8080.

docker run -d -p 9090:80 httpd

docker ps

The Apache default webpage was successfully accessed through the browser.

```
kavip@KaviPrakash:~$ docker run -d --name apache_server -p 9090:80 httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb700ef54: Pull complete
30973b009bab: Pull complete
85b0d820aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a390514d6767e8c62f29375d0577190be448f63b24f5f11d6b03f7bf18
Status: Downloaded newer image for httpd:latest
6a1b33988452b0f0e5ec0f143448d2c51a730335dfe7e1b171f951c727edd9a5
kavip@KaviPrakash:~$ docker ps
CONTAINER ID   IMAGE     COMMAND              CREATED         STATUS         PORTS                                     NAMES
6a1b33988452   httpd     "httpd-foreground"   4 seconds ago   Up 3 seconds   0.0.0.0:9090->80/tcp, [::]:9090->80/tcp   apache_server
kavip@KaviPrakash:~$
```



Step 6 – View Docker Containers

All containers were listed to verify their status.

docker ps -a

This displayed running and stopped containers.

```
kavip@KaviPrakash:~$ docker ps -a
CONTAINER ID   IMAGE              COMMAND                  CREATED         STATUS                       PORTS
                NAMES
37f5c706a7fc   python:3.11-slim   "bash"                   2 minutes ago   Exited (2) About a minute ago
                py_dev
6a1b33988452   httpd              "httpd-foreground"       3 minutes ago   Up 3 minutes                 0.0.0.0:9090->80/tcp, [::]
:9090->80/tcp   apache_server
7e91307e5ac4   alpine             "sh"                     6 minutes ago   Exited (130) 3 minutes ago
                alpine_test
efd2ac746b10   busybox            "echo 'Docker is wor…"   6 minutes ago   Exited (0) 6 minutes ago
                boring_moser
f711f5735469   busybox            "echo 'Docker is wor…"   8 minutes ago   Exited (0) 8 minutes ago
                nifty_margulis
kavip@KaviPrakash:~$
```

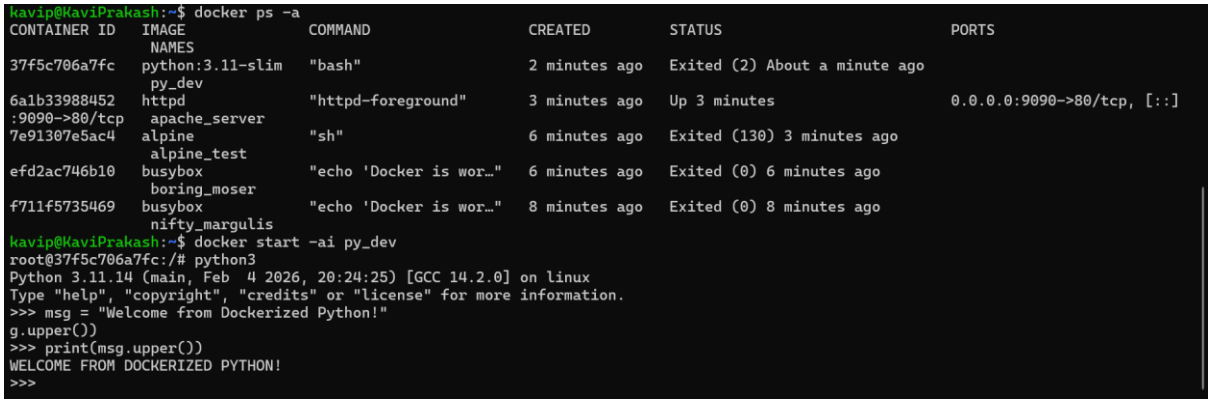
Step 7 – Run Python Container

A Python development container was launched.

docker run -it python:3.11-slim bash

python3

Python executed successfully inside the container.



Output

- **Docker Engine installed successfully.**
- **Containers were created and executed without errors.**
- **Apache web server ran successfully on localhost.**
- **Python environment functioned correctly inside Docker.**

Result

Docker was successfully installed and configured in WSL. Multiple containers were created and managed, and a Python runtime environment was executed inside a Docker container, demonstrating effective containerization.